

ABSTRACT

Our project is an Arduino-based interactive memory game designed to enhance cognitive skills such as memory and concentration. The game works by flashing lights in a random sequence, which players must correctly replicate by entering corresponding commands. As the player progresses, the sequences become increasingly complex, making the game progressively more challenging. The objective is to improve memory retention and pattern recognition through engaging and interactive gameplay.

INTRODUCTION

Memory games have long been recognized for their ability to enhance cognitive skills such as attention, concentration, and pattern recognition. In recent years, interactive, light-based memory games have gained popularity due to their engaging and immersive nature. These games not only provide entertainment but also serve as a valuable tool for exercising short-term memory in users of all ages.

Despite the availability of various memory games in the digital market, most of these applications are screen-based and lack tactile, hands-on interaction that can further stimulate cognitive processes. Traditional digital formats often fail to capture the physical engagement that makes memory games more dynamic and enjoyable. Recognizing this gap, we propose the development of an Arduino-based memory game to deliver a more interactive and engaging experience.

The objective of this project is to create a memory game where lights flash in a random order, requiring players to replicate the pattern by entering corresponding commands. Each correct input earns the player points, while mistakes lead to a reset or point deduction, depending on the game's design. The game will utilize LEDs to display the sequences and buttons or serial commands for player input, providing a straightforward yet challenging activity that stimulates memory and reaction time.

By leveraging the Arduino platform, the project ensures accessibility, cost-effectiveness, and potential scalability. The design will incorporate clear documentation and code structure, allowing future modifications for varying difficulty levels or expanded gameplay features. Ultimately, this project aims to deliver a fun, educational, and engaging experience while demonstrating the practical applications of Arduino in interactive game design.

LITERATURE REVIEW

Memory games have long been utilized as tools to enhance cognitive functions, particularly memory retention and recall. With advancements in technology, these games have transitioned from traditional formats to digital and tangible interactive designs, often leveraging platforms like Arduino to create engaging learning experiences.

Take the classic Simon game (**fig.1**) for example, it has been fundamental in studying working memory. These games involve recalling and repeating a sequence of lights, sounds, or vibrations, testing the player's ability to store and retrieve information. Research has demonstrated that such games can enhance cognitive abilities and increase engagement with tasks (Jones et al., 2015).



fig.1 – Simon memory game

Furthermore, Arduino in Cognitive Games (**fig.2**): Arduino has been widely used in educational projects, providing a simple yet powerful platform for developing interactive games. In projects related to memory games, Arduino enables easy interfacing with LEDs, buzzers, and vibration motors, making it an ideal tool for developing a responsive and interactive game system. [1]

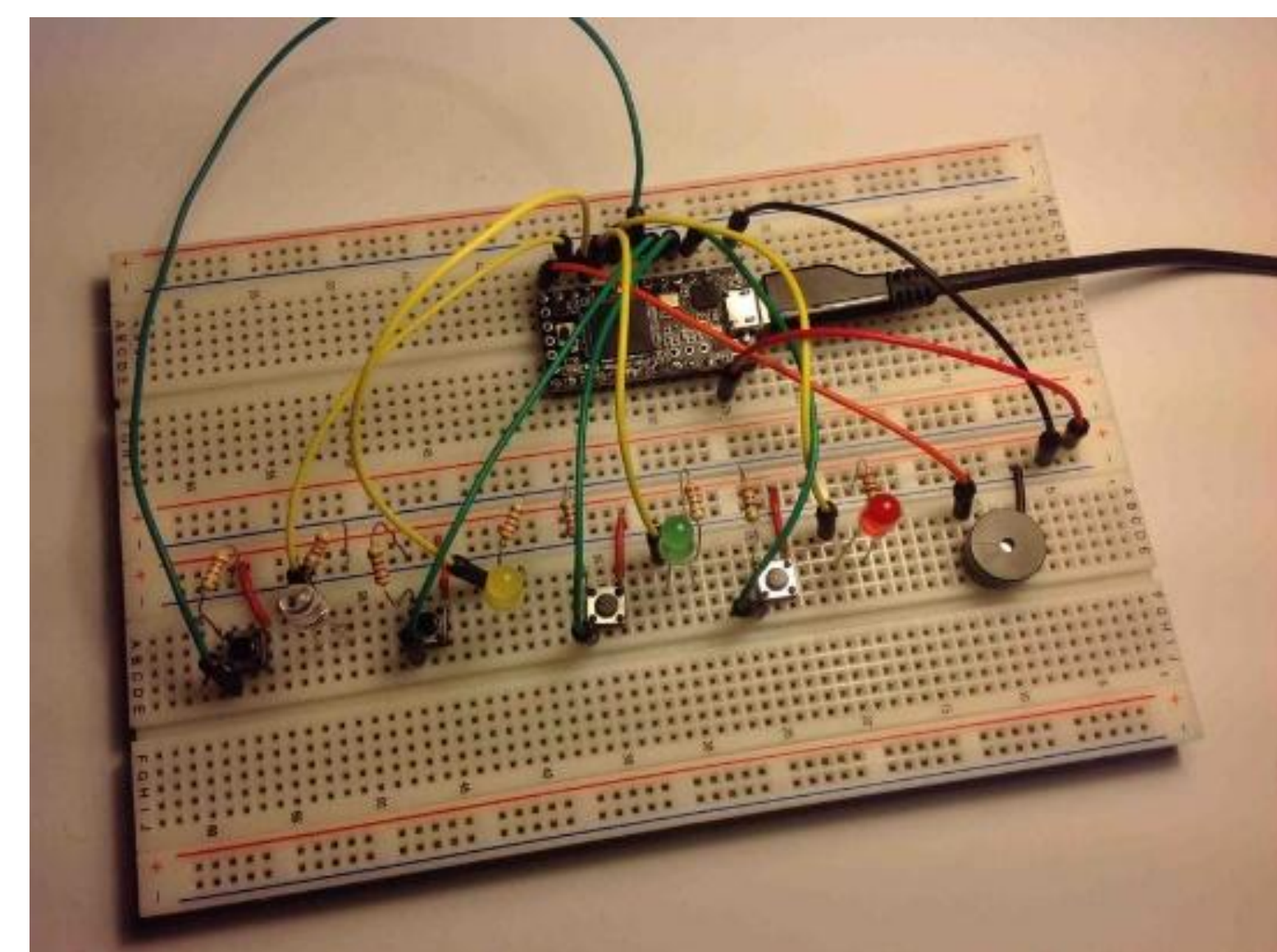
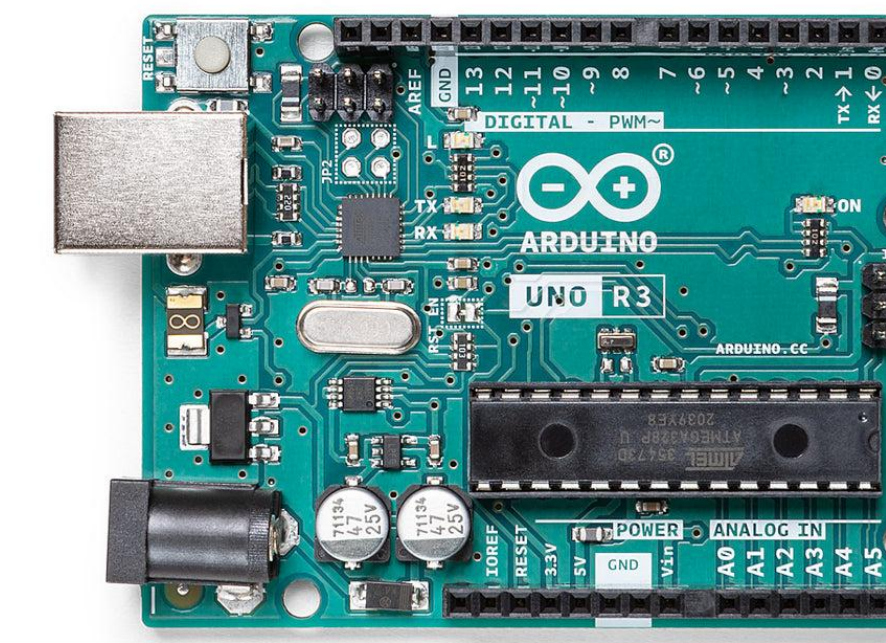


fig.2 – Teensy Memory Game

METHODOLOGY

The memory game was developed using

- an Arduino Uno board



- four LEDs (red, green, yellow, and blue)



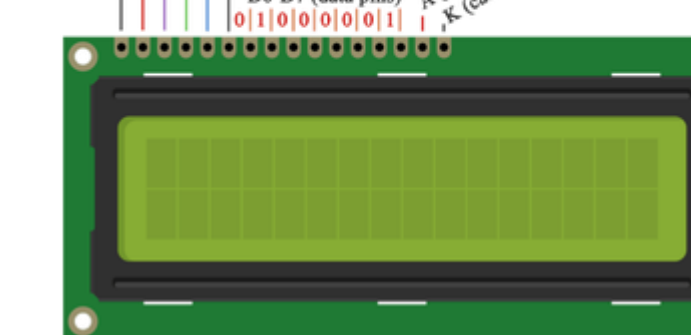
- four push buttons



- a piezo buzzer



- a 16x2 I2C LCD display.



- a 3D-printed model

Each button corresponds to an LED and a specific sound tone, enabling users to receive both visual and auditory feedback. The game logic is programmed in C++ within the Arduino IDE, employing functions to handle user input, sequence generation, feedback output, and game state transitions.

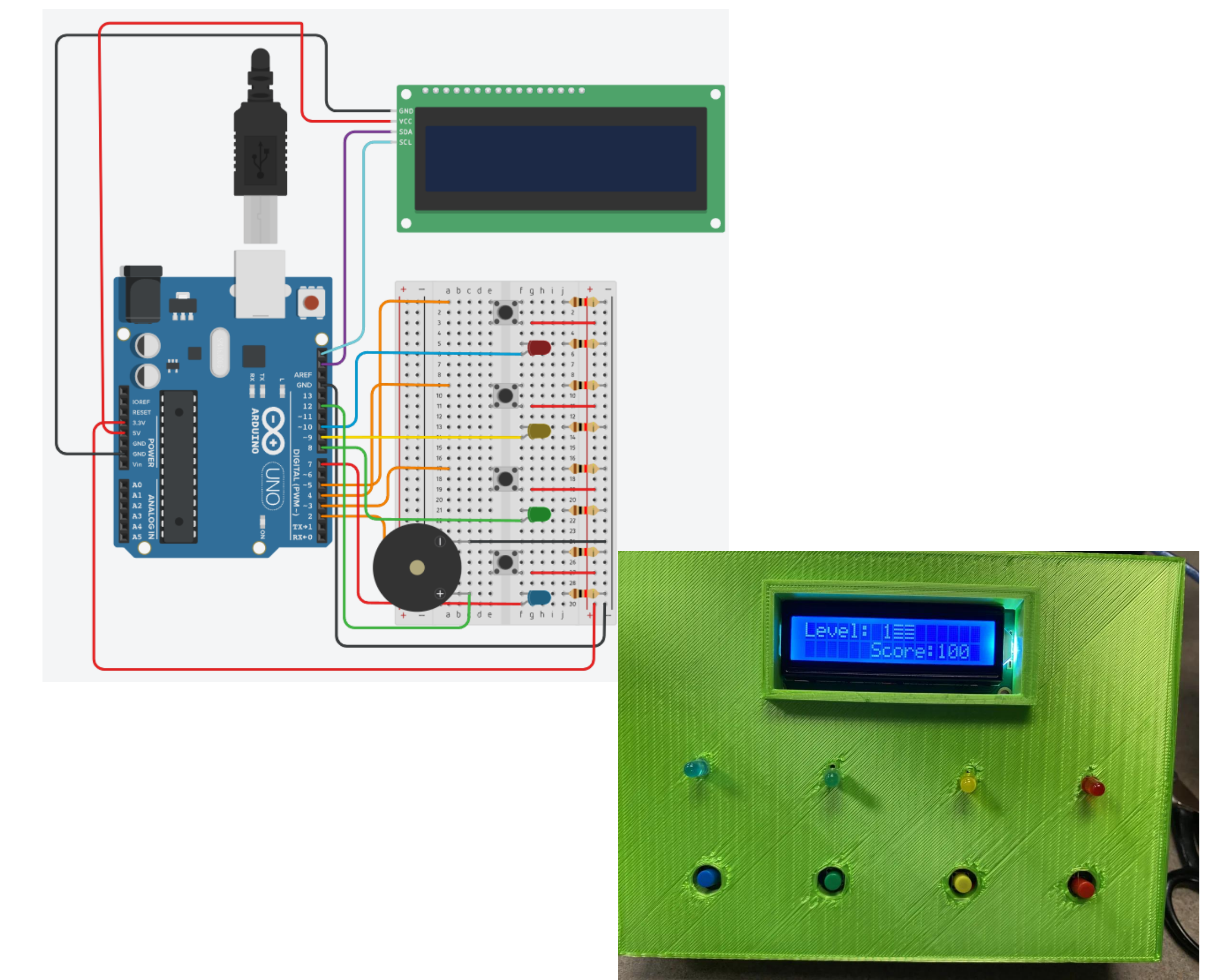
At startup, the game presents a menu with four options in the Serial Monitor: start Peaceful Mode, start Nightmare Mode, view top scores, or reset scores. In Peaceful Mode, sequences increase by one element per level and offer a moderate pace. Nightmare Mode introduces variability by increasing the level more aggressively and reducing the response time between actions, making it ideal for players seeking a greater challenge.

Each correct response progresses the player to the next level, while a wrong input ends the game and updates the high scores. Scores are tracked internally and displayed on the LCD. Sound cues from the buzzer enhance player awareness during interactions and signify success or failure. The game's modular design makes it easy to modify or expand in future iterations, such as adding more buttons or wireless functionality.

CONCLUSION

The Arduino Memory Game successfully illustrates how embedded systems and basic electronic components can be integrated to create a fun, interactive, and educational experience. Through features such as real-time scoring, adjustable difficulty, and user feedback via sound and LCD display, the game challenges users to strengthen their memory and concentration in an engaging way. Its smooth combination of hardware and software makes it an excellent example of beginner-to-intermediate Arduino development, particularly in educational settings.

Although the current version is fully functional and user-friendly, there is room for future enhancements, such as customizable game modes, multiplayer options, or Bluetooth functionality for mobile score tracking. Completing this project significantly deepened our understanding of programming, circuit design, and system integration. More importantly, it emphasized the importance of hands-on projects in developing critical thinking and problem-solving skills as emerging computer science students.



REFERENCES

1. Instructables. "Arduino Simple Memory Game." *Instructables*, 22 July 2020, www.instructables.com/Arduino-Simple-Memory-Game/?utm
2. P. Duch and T. Jaworski, "Enriching Computer Science Programming Classes with Arduino Game Development," *2018 11th International Conference on Human System Interaction (HSI)*, Gdansk, Poland, 2018, pp. 148-154, doi: 10.1109/HSI.2018.8430994.